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18 June 1986
SUPERSEDING
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(See 6.7)

MILITARY SPECIFICATION

SNUBBERS, FLUID PRESSURE, INSTRUMENT PROTECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. A snubber covered by this specification is a pressure transmission device which restricts the rate of fluid flow to a pressure sensing instrument and, as a result the rate of pressure change.

1.2 Classification. Snubbers covered by this specification shall be classified according to the following:

1	-	L	-	a	-	P	-	G
Type (see 1.2.1)		Class (see 1.2.2)		Composition (see 1.2.3)		End connection (see 1.2.4)		Cleanliness (see 1.2.5)

1.2.1 Types. The type of service in which the snubber will be used shall be designated by one of the following numbers:

Type	Service
1	Oil above 225 saybolt universal seconds (SUS) at operating temperatures.
2	Gasoline, fresh water, steam, light oils below 225 SUS.
3	Gases
4	Seawater (composition b only)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6685

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1.2.2 Class. The range of operating pressures shall be designated by the following letters:

Class	Operating pressure range
L	10 to 1000 pounds per square inch gauge (lb/in^2)
H	1001 to 6000 lb/in^2

1.2.3 Composition. The material composition of the snubber shall be designated by one of the following letters:

Composition	Material
a	Copper nickel alloy
b	Nickel-copper (monel) alloy
c	Corrosion resisting steel

1.2.4 End connection. The configuration for the snubbers end connection shall be designated by one of the following letters (see 3.3.2):

End connection	Application
P	Pressure gauge general applications (see 3.3.2.1.1).
T	Pressure transducer general applications (see 3.3.2.1.2).
O	Pressure gauge oxygen applications (see 3.3.2.2.1).
S	Other applications (see 6.1.1.3). The end connections must be specified in the contract ordering data.

1.2.5 Cleanliness. The method for cleaning the snubber shall be designated by one of the following letters:

Cleanliness	Application
G	General applications
X	Oxygen and dry nitrogen applications

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

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SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting.
UU-P-268 - Paper, Kraft, Wrapping.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes: Paperboard, Metal-Edged and Components.
PPP-B-676 - Boxes Setup.
PPP-C-850 - Cushioning Material, Polystrene Expanded, Resilient (For Packaging Uses).
PPP-C-1120 - Cushioning Material, Uncompressed Bound Fiber For Packaging.

MILITARY

MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-R-6130 - Rubber, Cellular, Chemically Blown.
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings.
MIL-N-17163 - Nickel-Copper Alloy, Wrought; (55-60 Percent Nickel) Low Permeability.
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-F-18866 - Fittings, Hydraulic Tube, Flared, 37 Degree and Flareless: Steel.
MIL-R-20092 - Rubber Sheets and Assembled and Molded Shapes, Cellular, Synthetic, Open Cell (Foamed Latex).
MIL-P-24212 - Pressure Transducer Equipment (Electrical).
MIL-V-24578 - Valves, Globe, Pressure Instrument, Stem Test Connection, Union End.
MIL-P-26514 - Polyurethane Foam, Rigid or Flexible, For Packaging.
MIL-T-28800 - Test Equipment For Use With Electrical and Electronic Equipment, General Specification For.
MIL-R-83248 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels in Ships of the United States Navy.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-1330 - Cleaning and Testing of Shipboard Oxygen, Nitrogen and Hydrogen Gas Piping Systems.
MIL-STD-1622 - Cleaning of Shipboard Compressed Air Systems.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fittings, Standard Dimensions for.

2.1.2 Other Government drawing. The following other Government drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 803-1385850 - Piping, Instrument, Pressure, For All Service.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D 3951 - Standard Practice for Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The snubbers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2.1 and 6.3).

3.2 Material.

3.2.1 Snubber body. The material for the snubber body and internal metal parts (see 3.3.1) shall be in accordance with 3.2.1.1, 3.2.1.2 and 3.2.1.3 for compositions a, b and c respectively.

3.2.1.1 Composition a. The material for composition "a" shall be copper-nickel alloy in accordance with MIL-C-15726.

3.2.1.2 Composition b. The material for composition "b" shall be nickel-copper (monel) alloy in accordance with MIL-N-17163 or QQ-N-281. Composition b is the only metallic material that is certified for seawater (type 4) applications.

3.2.1.3 Composition c. The material for composition "c" shall be corrosion-resisting steel in accordance with QQ-S-763, classes 304L, 316L, 321, or 347. Where the end connections (see 3.3.2) do not consist of nipples or connections which are meant to be welded, then classes 304 and 316 are also acceptable.

3.2.2 Nonmetallic materials. Nonmetals shall be moisture and flame resistant, shall not support fungus growth, and shall not be adversely affected by the ambient environments specified in the configuration and performance requirements of this specification. Nonmetals shall not chemically react, degrade, or outgas when subjected to the following: air, distilled water, sea water, salt, petroleum and silicone based oils, oil solvents, prolonged (greater than one month) periods of exposure to ambient temperatures, prolonged periods of exposure to elevated temperatures (up to 400°F), and exposure to prolonged cycling periods from ambient to elevated temperatures. Any material which provides a nutrient medium for fungus and insects shall not be used. Nonmetals shall be fungus inert materials in accordance with requirement 4 of MIL-STD-454.

3.2.3 Pipe nipple. When specified for use (see 3.3), the material for pipe nipples shall be nickel-copper (monel) alloy in accordance with MIL-N-17163.

3.2.4 O-ring. When specified for use (see 3.3.2.1.1), the material for O-rings shall be fluorocarbon rubber in accordance with MIL-R-83248, type I.

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3.2.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.6 Prohibited materials. The snubber shall not contain a prohibited material as specified in MIL-T-28800.

3.2.6.1 Type 3. Snubbers with a porous metal composition (internal parts and configuration) shall not be certified for gas (type 3) applications.

3.2.6.2 Type 4. Snubbers with a porous metal composition (internal parts and configuration) shall not be certified for seawater (type 4) applications.

3.3 Configuration. The snubber shall consist of no more than two primary components except as provided below.

(1) Internal parts needed to perform spike suppression and fluctuation/pulsation dampening.
(2) Snubbers used in oxygen applications may have pipe nipples welded to the snubber body.

3.3.1 Snubber body. The length of the body of the snubbers when assembled shall not exceed 4 inches. The body shall have hexagonal cross-sections of sufficient widths to allow the use of a wrench for installing and removing. The distance across flats shall be not less than 0.375 inch nor greater than 1.25 inches.

3.3.1.1 Positive retaining feature. Internal parts shall be positively retained in place, preventing them from becoming loose when the snubber is assembled. Split pin type retaining pins shall not be used.

3.3.2 End connections.

3.3.2.1 General applications.

3.3.2.1.1 Pressure gauge applications. Snubbers used for pressure gauge applications shall have 9/16-18 UNF-3A external threads on the inlet connection and 9/16-18 UNF-3B internal threads on the outlet connection in accordance with Drawing 803-1385850. When specified, each snubber shall be furnished with a union nut, tailpiece and O-ring (see 6.2.1).

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3.3.2.1.2 Pressure transducer applications. Snubbers used for pressure transducer applications shall have a 7/16-20 UNF flareless tube connection (1/4 inch outside diameter (od)) in accordance with MIL-F-18866 on the inlet connection. The outlet connection shall be 7/16-20 UNF-2A external thread to mate with the MS16142 female connection port of the pressure transducer. An O-ring shall be supplied on the outlet connection.

3.3.2.2 Oxygen applications.

3.3.2.2.1 Pressure gauge applications. Snubbers used for oxygen systems shall have a copper-nickel pipe nipple (see 3.2.3) as the inlet connection which is welded to the snubbers body in accordance with MIL-STD-278. The pipe nipple shall be 0.25 inch nominal pipe size (nps), 0.120 inch minimum wall thickness, and 6 inches long. The outlet connection shall be a socket weld connection sized to accept a 0.25 inch nps pipe nipple as described above.

3.3.3 Cleaning. The snubber parts that come into contact with the service fluid shall be free of all loose scale, rust, grit, fillings, and free of mercury, calibration liquids, oil, grease, solvents, and other organic materials.

3.3.4 Welding and brazing. When applicable, snubber parts shall be joined by welding. Only materials which are considered to be readily weldable and of acceptable thickness shall be used. Welding shall be in accordance with MIL-STD-278.

3.3.5 Marking. Each snubber shall be metal stamped or otherwise permanently marked with the maximum working pressure, type, class, composition, and manufacturer's stamp or trademark. The maximum working pressure shall be followed by the abbreviation "lb/in^2", for example 6000 lb/in^2. The type, class, and composition shall be marked for type, class and composition, for example 1-H-b for type 1, class H, composition b.

3.3.5.1 Flow direction. The direction of flow through the snubbers shall be indicated by an arrow with the word gauge at the point of the arrow. The arrow shall be raised from the surface in a permanent manner and to the extent that it can be easily identified by human touch. The word gauge shall be metal stamped or otherwise permanently marked.

3.4 Technical data. Contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2).

3.5 Performance.

3.5.1 Steady state transmission. When tested in accordance with the procedure specified in 4.5.1, the difference in the pressure transducer calibration performed with and without the snubber under test shall not be greater than plus or minus 0.5 percent of the pressure transducer span.

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3.5.2 Dampening. When tested in accordance with the procedure specified in 4.5.2, the test pressure transducer shall exhibit an output of at least 90 percent of the applicable steady state pressure step within the time interval 0.5 seconds to 2.5 seconds after the initiation of the pressure step change. In addition, the output of the test transducer shall not overshoot the final steady state pressure by more than 0.1 percent of the span of the pressure step.

3.5.3 Spike suppression. When tested in accordance with the procedure specified in 4.5.3, the output of the test pressure transducer shall not deviate from its steady state value by more than 0.5 percent of the span of the applied pressure spike.

3.5.4 Temperature. The snubbers under test shall not show any indication of leakage or damage as a result of temperature testing specified in 4.5.4. After any of the temperature tests specified in 4.5.4, the snubber shall be capable of meeting the dampening test requirements of 3.5.2. For the high and low temperature tests the time interval shall be 0.25 to 2.75 seconds.

3.5.5 Pressure integrity. The snubbers under test shall be capable of withstanding the pressure specified in 4.5.5 without showing any evidence of leakage or damage. Upon completion of the test, the snubber shall meet the dampening test requirements of 3.5.2.

3.5.6 Vibration. The snubber under test shall show no indication of leakage or damage as a result of vibration testing specified in 4.5.6. Upon completion of the test, the snubber shall meet the dampening test requirements of 3.5.2.

3.5.7 Shock. The snubber under test shall not show any indication of leakage or damage as a result of shock testing specified in 4.5.7. Upon completion of the test, the snubber shall meet the dampening test requirements of 3.5.2.

3.5.8 Wear. The snubber under test shall be capable of withstanding the pressure transients specified in 4.5.8 without indication of leakage or damage. Upon completion of the test, the snubber shall be capable of meeting the dampening test requirements of 3.5.2, except that the minimum time to reach 90 percent of the steady state pressure shall be decreased from 0.5 to 0.25 seconds.

3.5.9 Burst pressure. The snubber under test shall show no indication of leakage or damage as a result of the burst pressure test specified in 4.5.9. Upon completion of the test, the snubber shall meet the dampening test requirements of 3.5.2 except that the minimum time to reach 90 percent of the steady state pressure shall be decreased from 0.5 to 0.25 seconds.

3.6 Cleaning and surface finishes. Surfaces of castings, forgings, molded parts, stampings, machined, and welded parts shall be free of defects such as cracks, porosity, undercuts, voids, and gaps as well as sand, dirt, fins, sharp edges, scale, flux, and other harmful or extraneous materials. External surface shall be smooth and edges shall be either rounded or beveled. There shall be no burn-through. There shall be no warpage or dimensional change due to heat from welding operation. There shall be no damage to adjacent parts resulting from the welding.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable to the Government for the performance of the inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection system. The contractor shall provide and maintain an inspection system and a calibration system in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.1.2.1 Calibration system requirements. The contractor shall provide and maintain a calibration system for test equipment.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the Government. Qualification inspection shall consist of the examination and tests specified in table I and in the sequence listed.

4.2.1.1 Sample size. Two snubber samples of each type, class and composition shall be subjected to qualification inspection.

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TABLE I. Inspection, examination, and tests.

Inspection	Requirement	Test procedures	Qualification inspection	Quality conformance inspection
Examination	3.2, 3.3, 3.6	4.4	X	X
Steady state transmission	3.5.1	4.5.1	X	X
Dampening	3.5.2	4.5.2	X	X
Spike suppression	3.5.3	4.5.3	X	-
Temperature	3.5.4	4.5.4	X	-
Pressure integrity	3.5.5	4.5.5	X	-
Vibration	3.5.6	4.5.6	X	-
Shock	3.5.7	4.5.7	X	-
Wear	3.5.8	4.5.8	X	-
Burst pressure	3.5.9	4.5.9	X	-

4.2.1.2 Retention of qualification. To retain qualification, the contractor shall forward a retention of qualification report to NAVSEA at 2 year intervals. The report shall consist of a summary of quality conformance inspection results of snubbers for delivery, indicating as a minimum, the number of lots that have passed, the number that have failed, and the designation of the lots which have failed. The results of inspections of all reworked lots shall be accounted for and identified. If the summary of the inspection results indicates nonconformance with specification requirements, and corrective actions acceptable to NAVSEA have not been taken, action may be taken to remove the snubbers from the qualified products list.

4.2.1.2.1 Failure to submit the report within 30 days after the end of each 2 year period may result in loss of qualification for the snubber. In addition to the periodic submission of inspection data, the contractor shall immediately notify NAVSEA at any time during the 2 year period that the inspection data indicates failure of the qualified snubber to meet the requirements of this inspection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed at the place of manufacture (see 4.2.2.2). Quality conformance inspection shall consist of the examination and tests specified in table I and shall be witnessed by a Government inspector.

4.2.2.1 Lot. A lot shall consist of all snubbers of the same type, class, and composition manufactured and offered for delivery at one time.

4.2.2.2 Sampling for quality conformance inspection. A random sample of snubbers shall be selected by a Government inspector from each lot of snubbers (see 4.2.2.1). The selection shall be performed in accordance with MIL-STD-105, inspection level I, acceptable quality level (AQL); 1-1/2 percent defects shall apply.

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4.2.2.3 Acceptance criteria. If any snubber sample exhibits a critical or major defect listed in table II, no snubbers shall be accepted for quality conformance inspection until the contractor has determined the cause of the defect and has taken the necessary action to correct or eliminate the defect from each snubber in the lot. The failed test shall be repeated to demonstrate that the corrective action will enable the snubbers to conform to the requirements of this specification. In addition, the results of previous tests may be deemed invalid, unless the contractor can prove to the satisfaction of the Government that such tests would not be adversely impacted by the corrective action. If the number of snubbers that contain a minor defect exceeds the acceptable quantity listed in table II, no snubber shall be accepted for quality conformance inspection until all snubbers in the lot are inspected for the defect and necessary corrective action taken to enable all snubbers in the lot to conform to the requirements of this specification.

TABLE II. Classification of defects.

Categories	Defects	Requirements	AQL
Critical			
1	Evidence that snubbers for oxygen service are not cleaned, improperly marked, or improperly packaged	3.3.3, 5.1.1.1	0
2	Snubber does not meet the pressure integrity test (see 4.5.5)	3.5.5	0
Major			
101	Evidence of unauthorized material	3.2	0
102	Inspection system not provided, dimensional tolerances not maintained, mounting dimensions not interchangeable, pressure connection wrong design, and dimensions erroneous	3.3, 3.3.2, 4.4	0
103	Snubber does not meet steady-state transmission or dampening tests (see 4.5.1 and 4.5.2)	3.5.1, 3.5.2	0
104	Markings not provided or erroneous; part number not provided	3.3.5	0

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TABLE II. Classification of defects. - Continued

Categories	Defects	Requirements	AQL
Minor			
202	Evidence that snubbers for nonoxygen service are not cleaned	3.3.3	1-1/2
203	Finish non-conforming, scratched, chipped; burrs not removed, color non-conforming	3.6	4
204	Cleaning, drying, preservation, packaging, packing and markings non-conforming	5.	4

4.3 Quality conformance inspection reports. Quality conformance inspection reports shall be prepared in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.4 General examination. The snubbers shall be subjected to a thorough examination to ascertain that the material, finish, workmanship, construction, assembly, dimensions, and marking are in conformance with the requirements of this specification. Examinations shall be limited to disassembling the snubbers to the intent that the performance, durability, or appearance would not be affected.

4.4.1 Quality conformance inspection examination. The snubbers selected in accordance with 4.2.2.2 shall be examined to determine conformance to the requirements of this specification, and the classification of defects in table II. The acceptable quality level for each defect shall be as shown in table II in accordance with MIL-STD-105.

4.4.1.1 Dimensional verification. Snubber samples selected in accordance with 4.2.2.2 shall be subjected to dimensional verifications of the pressure connection threaded end to verify conformance to Drawing 803-1385850, or the applicable document for the type threaded end or connection specified. In order to comply with the dimensional verification, all dimensions, concentricities, and perpendicularities affecting interchangeability of parts, sealing effectiveness and strength, shall be measured with sufficient precision to verify conformance to the applicable document. For a snubber having an O-ring union pressure connection, samples of the tail piece and union nut shall also be subjected to this dimensional verification. An AQL of 4 percent defect shall apply for the snubber's threaded ends. For the tail piece and union nut, inspection level II AQL 1-0 percent defects shall apply, when the pressure connection is the O-ring union type.

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4.5 Test procedures. Tests shall be conducted as described below. Except for the pressure integrity test, class L tests shall be performed using 0/1000 lb/in^2 pressure transducers and class H tests shall be performed using 0/1000 lb/in^2 pressure transducers. Pressure transducers used shall meet or exceed the accuracy, temperature and safety (insulation resistance, overpressure and burst pressure) requirements of MIL-P-24212. A storage oscilloscope or high frequency (greater than 1000 Hz) oscillographic recorder shall be used to monitor the pressure transducer outputs. Pressure transducer outputs that result from the steady state transmission tests shall be monitored by a digital or analog multimeter. The pressure integrity test and burst pressure test shall be monitored with either a pressure transducer or safety solid front pressure gauge of suitable range which meets the accuracy and safety requirements specified above. Except for the wear test, which shall use air or nitrogen for all types, the fill fluid in the snubber shall be water for types 2 and 4, air or nitrogen for type 3 and 2190-TEP oil (in accordance with MIL-L-17331) for type 1. An equivalent oil, in accordance with MIL-L-17331, may be substituted for 2190-TEP oil. Unless otherwise indicated, the pressure transducer shall be connected to the output side of the snubber, with only an adapter fitting between them. Except where the following factors are the variables, the tests in this specification shall be conducted with the equipment and instrumentation operating under the following conditions:

(a) Ambient temperature shall be 75 $\pm$ 10°F.
(b) Relative humidity shall be 50 $\pm$ 10 percent.
(c) Supply voltage shall be 115 $\pm$ 5 volts.
(d) Supply frequency shall be 60 $\pm$ 2 Hz.

4.5.1 Steady state transmission. The pressure transducer shall be calibrated with and without a snubber. The calibration shall be done over five equally spaced intervals, both upscale and downscale, for the applicable class (see 1.2) as shown in table III. The snubber shall meet the requirements of 3.5.1.

TABLE III. Steady state transmission pressures.

Snubber class (see 1.2.2)	Snubber range (lb/in^2)	Test pressures (lb/in^2)
L	10/1000	10, 250, 500, 750, 1000
H	1001/6000	1200, 2400, 3600, 4800, 6000

4.5.2 Dampening. The dampening test shall be conducted utilizing an apparatus similar to that shown on figure 1. A step change in pressure from ambient to 85 $\pm$ 5 percent of the upper value of the snubber class operating range shall be applied to the snubber under test. The pressure step change shall be accomplished in 50 milliseconds or less. Both the test and monitoring pressure transducers utilized to record the step response shall have a damped natural frequency of at least 500 Hz. This test shall also be known as the reference measurement when performed at the end of another test. The snubber shall meet the requirements specified in 3.5.2.

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4.5.2.1 Procedure for dampening test. The following procedure shall be followed to perform the dampening test.

(a) Ensure that valves V2, V6, V8, V9 and V10 are closed and that valves V1, V3 and V4 are open.
(b) Utilizing valves V3, V4 and V5; fill the snubber/transducer test section with fluid. The fluid type shall be appropriate for the snubber under test. Close valves V3, V4 and V5.
(c) Utilizing valve V1, pressurize the high pressure reservoir with nitrogen to a value sufficient to ensure that the pressure step change in the test snubber/transducer test section meets the test conditions of 4.5.2. Close valve V1.
(d) Actuate the quick opening valve, V2, and record the output of both pressure transducers.
(e) After steady state has been achieved, open valve V9 to equalize the system pressure.
(f) Utilizing valve V10, vent the system to atmosphere, ensuring that any test fluid is confined in a suitable reservoir.
(g) If required, purge the entire system with nitrogen to remove residual traces of the test fluid.

4.5.3 Spike suppression. The spike suppression test shall be constructed utilizing an apparatus similar to that shown on figure 1. A pressure spike, having an amplitude of 85 $\pm$ 5 percent of the upper value of the snubber class operating range, shall be applied to the snubber under test. The duration of the pressure spike as measured between the 10 percent points (see figure 2) shall be not less than 25 milliseconds nor more than 50 milliseconds. The pressure transducers utilized to record both the dampened and undampened pressure spike shall have a damped natural frequency of at least 500 Hz. The snubber shall meet the requirements as specified in 3.5.3.

4.5.3.1 Procedure for spike suppression test. The following procedure shall be followed to perform the spike suppression test.

(a) Ensure that a rupture disc with a burst rating equal to 90 percent of the upper value of the snubber class operating range is installed in the rupture disc assembly.
(b) Ensure that valves V2, V9 and V10 are closed and that valves V3, V4, V5, V6, V7 and V8 are open.
(c) Utilizing valves V3, V4, V5 and V7; fill the snubber/transducer test section with the appropriate fluid type for the snubber under test. Close valves V3, V4, V5 and V7.
(d) Utilizing valve V1, pressurize the high pressure reservoir to a value sufficient to ensure a pressure spike amplitude in the snubber/transducer test section meets the test conditions of 4.5.3. Close valve V1.
(e) Actuate quick opening valve V2 and record the pressure transducers output.
(f) After steady state has been achieved, vent the system to atmosphere utilizing valve V10, ensuring that any test liquid is confined in a suitable reservoir.
(g) If required, purge the entire system with nitrogen to remove residual traces of the test fluid.

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4.5.4 Temperature.

4.5.4.1 High temperature. Place the snubber in a temperature test chamber. Pressurize the snubber to 500 ± 5 lb/in^2 for a class L snubber, and 3000 ± 30 lb/in^2 for a class H snubber. Increase the temperature to 145 ± 5°F and hold. After the temperature has stabilized, the snubber shall remain in the temperature test chamber for at least 4 more hours while pressurized. Perform a reference measurement after this 4 hour period, with the snubber remaining at high temperature. Remove the snubber from the temperature test chamber. Allow it to stabilize to ambient for at least 4 hours and perform a second reference measurement.

4.5.4.2 Low temperature. Place the snubber in a temperature test chamber. Pressurize the snubber to 500 ± 5 lb/in^2 for a class L snubber, and 3000 ± 30 lb/in^2 for a class H snubber. Decrease the temperature to 40 ± 5°F and hold. After the temperature has stabilized, the snubber shall remain in the temperature test chamber for at least 4 more hours while pressurized. Perform a reference measurement after this 4 hour period, with the snubber remaining at low temperature. Remove the snubber from the temperature test chamber. Allow it to stabilize to ambient for at least 4 hours and perform a second reference measurement.

4.5.4.3 Storage temperature. The snubber shall be placed in a temperature test chamber. The snubber shall be unpressurized. The test shall consist of five complete temperature cycles, as shown in table IV. After completion of the fifth cycle, the snubber shall be removed from the temperature test chamber and allowed to stabilize at ambient conditions for at least 4 hours. A reference measurement shall then be performed.

TABLE IV. Storage temperature cycle.

Step	Temperature condition	Step duration
1	Increase temperature 1/ from 75 ± 5°F to 145 ± 5°F	1 hour maximum
2	Constant temperature of 145 ± 5°F	22 hours minimum
3	Decreasing temperature from 145 ± 5°F to -10 ± 5°F	2 hours maximum
4	Constant temperature of -10 ± 5°F	6 hours minimum
5	Increasing temperature from -10 ± 5°F to 75 ± 5°F	1 hour maximum

1/ Ambient temperature for the first cycle.

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4.5.5 Pressure integrity. A pressure source is required, with a valve to isolate the source once the pressure is at the specified condition. The snubbers shall be subjected to the following applicable tests for 1 hour as shown in table V. A reference measurement shall be performed at the end of this test.

TABLE V. Pressure integrity test conditions.

Snubber class (see 1.2.2)	Pressure range (lb/in^2)	Test conditions (lb/in^2)
L	10/1000	1500
H	1001/6000	9000

4.5.6 Vibration. The vibration test consists of the exploratory test, the variable frequency test, and the endurance test. These tests shall be conducted in the sequence listed. Each of the three tests shall be conducted in each of the three mutually perpendicular axis. All three tests shall be completed in one axis before performing the tests in another axis.

(a) Exploratory test.

(1) Maintain each discrete frequency from 5 to 100 Hz at 1 Hz intervals for a minimum of 15 seconds, or a sweep rate that shall not exceed 4 Hz per minute.
(2) Displacements/accelerations shall be as specified in table VI.
(3) Frequencies and locations where resonance occur during this test shall be noted.

(b) Variable frequency test.

(1) Discrete frequency interval of 1 Hz.
(2) Frequency range 5 to 100 Hz.
(3) Each discrete frequency shall be maintained for a minimum of 5 minutes.
(4) Displacements shall be as specified in table VI.
(5) Frequencies and locations where resonance occurs during this test shall be noted.

(c) Endurance test. The snubbers shall be subjected to a 2 hour endurance run at each resonance.

(1) Displacments/accelerations shall be the variable frequency test values that are specified in table VI.
(2) If no resonance is found, a 2 hour endurance run shall be performed at 100 Hz.

A reference measurement shall be performed after the conclusion of the vibration test.

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TABLE VI. Vibratory displacement criteria.

Frequency range (Hz) (inclusive)	Table displacement (inches, peak to peak)	
	Exploratory test	Variable frequency test
5 to 20	0.020 ± 0.004	0.060 ± 0.012
21 to 50	0.010 ± 0.002	0.040 ± 0.008
51 to 100	0.005 ± 0.001	0.020 ± 0.004

4.5.6.1 Vibration fixture. The snubber shall be connected to the inlet side of an instrument cutoff valve. The valve shall be in accordance with MIL-V-24578. The valve and snubber assembly shall be mounted to a vertical panel of the vibration fixture, using the valve mounting nut and lockwasher. A flexible hose of suitable pressure rating shall be used to connect the pressure source to the inlet side of the snubber. The outlet side of the instrument cutoff valve shall be connected by a 0.25 inch nps pipe nipple of sufficient length to reach the top of the vibration fixture and shall be capped after filling the test piping. The pipe shall be clamped to the vibration fixture to prevent movement during testing. The snubber shall be pressurized to 500 ± 5 lb/in^2 for a class L snubber and to 3000 ± 30 lb/in^2 for a class H snubber.

4.5.7 Shock. The shock test shall be conducted in accordance with MIL-S-901 for grade A, class 1, type A equipment. A total of nine blows shall be applied, three blows applied parallel to each axis of the snubber. The three blows for each axis shall be conducted from hammer drop heights of 1, 3, and 5 feet. A reference measurement shall be performed after this test.

4.5.7.1 Shock fixture. The snubber shall be connected to the inlet side of an instrument cutoff valve. The valve shall be in accordance with MIL-V-24578. The valve and snubber assembly shall be mounted in the center of a type 6.D-1 adapter plate using the valve mounting nut and lockwasher. A flexible hose shall be used to connect the pressure source to the inlet side of the snubber. The outlet side of the instrument cutoff valve shall be connected to a pressure transducer. The hose shall be of sufficient length to allow mounting the pressure transducer on a fixture, not on the shock machine foundation. The snubber shall be pressurized to 500 ± 5 lb/in^2 for a class L snubber, and to 3000 ± 30 lb/in^2 for a class H snubber. The pressure transducer output shall be monitored during each hammer drop.

4.5.8 Wear. The wear test shall be conducted utilizing an apparatus similar to that shown on figure 3. A series of pressure spikes, each having an amplitude of 85 ± 5 percent of the upper value of the snubber class operating range, shall be applied to the snubber. Duration of each pressure spike as measured between the 10 percent points (see figure 2) shall not exceed 50 milliseconds. The snubber shall be subjected to at least 50,000 pressure spikes which shall be applied at a rate between 0.5 to 2.5 Hz. Air or nitrogen shall be utilized as the working fluid for all snubber fluid services. The test and monitoring pressure transducers shall be used for reference purposes only.

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4.5.9 Burst pressure test. The snubber shall be connected to a hydraulic pump using a water medium capable of generating a pressure 2-1/2 times the maximum operating pressure of the snubber under test. A shutoff valve shall be installed to isolate the snubber and test transducer or gauge from the pressure source. The outlet port of the snubber shall be plugged after ensuring all air has been vented from the test piping. The system shall be pressurized to a value equal to 2-1/2 times the maximum operating pressure of the snubber under test. The pressure shall be maintained for a period of 1 hour.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation. Preservation shall be level A, C or commercial, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying.

5.1.1.1.1 Types 1, 2 and 4. Unless otherwise specified (see 6.2.1), cleaning and drying of types 1, 2 and 4 snubbers shall be in accordance with MIL-STD-1622.

5.1.1.1.2 Type 3. Type 3 snubbers shall be cleaned and dried in accordance with MIL-STD-1330.

5.1.1.2 Unit protection. Preservatives shall not be used.

5.1.1.2.1 Bagging. Types 1, 2 and 4 snubbers shall be individually placed in a bag conforming to MIL-B-117 type I, class B, style 2 having a minimum thickness of 0.004 inch. Type 3 snubbers shall be individually placed in a bag conforming to MIL-B-117 type I, class E, style 2 having a minimum thickness of 0.006 inch. Snubber threads shall be protected by a plastic cap and to prevent puncture of the plastic bag. Any sharp edges and protrusions of the snubbers shall be cushioned before placing in the plastic bag.

5.1.1.2.2 Boxing. Each bagged snubber shall be individually unit packed in a box conforming to PPP-B-566 variety 2, process II, PPP-B-636 class weather-resistant, PPP-B-665 class 2 or PPP-B-676 with box selection and style at the contractor's option. Box closure and sealing shall conform to the requirements for water-proofing or weather-resistant as specified in the applicable box specification or appendix thereto.

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5.1.1.3 Intermediate packing. Snubbers of the same type, class and composition, unit packed as specified in 5.1.1.2.1 shall be intermediate packed in close-fitting fiberboard boxes conforming to PPP-B-636, type CF, class weather-resistant, style optional. Intermediate packs shall contain uniform quantities. Boxes shall be closed method V in accordnace with the appendix to the box specification.

5.1.2 Level C. Cleaning and drying, unit protection and boxing shall be as specified for level A (see 5.1.1) except that boxes may be of the non water-proof domestic class or variety.

5.1.3 Commercial. Type 1, 2 and 3 snubbers shall be individually packaged in accordance with ASTM D 3951. Type 3 snubbers shall be cleaned, dried and bagged as specified for level A (see 5.1.1) with boxing in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level B, C or commercial, as specified (see 6.2.1).

5.2.1 Level B. Snubbers preserved as specified in 5.1 shall be packed in fiberboard containers conforming to PPP-B-636 class weather-resistant or PPP-B-640 class 2 with container selection at the option of the contractor. Containers shall be closed, sealed and reinforced, with reinforcement utilizing non metallic banding or pressure sensitive reinforced tape. Box closure for PPP-B-636 containers shall conform to method V of the appendix to the box specification. Intermediate containers conforming to 5.1.1.3 herein, require no further overpacking.

5.2.2 Level C. Snubbers preserved as specified in 5.1 shall be packed as specified for level B (see 5.2.1) except that containers shall be of the non weather-resistant class. Containers shall be closed as specified in the appendix to the container specification with method I closure applicable to PPP-B-636 containers.

5.2.3 Commercial. Snubbers preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Cushioning, filler, dunnage and wrapping material.

5.3.1 Levels A, B and C. Use of excelsior, newspaper, shredded paper (all types, including wax paper) and similar hygroscopic or nonneutral materials and all types of loose-fill materials for applications such as cushioning, filler, stuffing and dunnage for materials destined for shipboard stowage and use is prohibited. Cushioning and wrapping materials selected shall have properties/characteristics for resistance to fire, examples are:

UU-P-268 - Paper, Kraft Wrapping Type II, Grade C or D
PPP-C-850 - Polystyrene, Expanded Grade SE, Type I or II Only
PPP-C-1120 - Bound Fiber, Uncompressed Type III or IV, Class C
MIL-R-6130 - Cellular Rubber Grade A
MIL-R-20092 - Cellular Rubber Class 1 or 4
MIL-P-26514 - Polyurethane Foam (Rigid or Flexible).

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5.3.2 Commercial preservation and packing. When loose fill type materials are used for preservation and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned with loose-fill material shall not be taken onboard ship. Remove and discard loose-fill material. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

5.4 Marking. In addition to any special marking required (see 6.2.1), interior unit packs and exterior shipping containers for levels A, B and C shall be marked in accordance with MIL-STD-129 including the bar-code markings specified therein. Type 3 unit packs (see 5.1.1.2.1) shall be marked with a warning label as specified in MIL-STD-1330 inserting "oxygen" or "nitrogen" and so forth, as applicable for the system use. The labels shall be green paper and shall match FED-STD-595, color no. 14187. Lettering shall be black.

6. NOTES

6.1 Intended use. Snubbers are intended for use in pressure instrument piping systems to suppress pressure spikes and dampen both pressure fluctuations and pulsations before they can reach and damage the pressure sensing instruments.

6.1.1 End connections.

6.1.1.1 Standard applications.

6.1.1.1.1 Pressure gauge applications. Snubbers intended for use on pressure gauges shall have 9/16-18 UNF-3A external threads on the inlet connection and 9/16-18 UNF-3B internal threads on the outlet connection. Snubbers shall be installed in accordance with Drawing 803-1385850.

6.1.1.1.2 Pressure transducer applications. Snubbers intended for use on pressure transducers shall have 7/16-20 UNF flareless tube (1/4 od) connection on the inlet connection and a 7/16-20 UNF-2A external thread on the outlet connection. The outlet of the snubber shall be threaded directly into the 7/16-20 UNF-2B connection port of the pressure transducer.

6.1.1.2 Oxygen applications.

6.1.1.2.1 Pressure gauge applications. Snubbers intended for use in oxygen systems shall have a 0.25 inch nps pipe nipple, 0.120 minimum wall thickness, 6 inches long on the inlet connection and a socket weld outlet connection sized to accept 0.25 nps pipe. The snubber shall be welded into the piping system in accordance with the requirements of MIL-STD-278.

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6.1.1.3 Other applications. Snubbers with end connections other than specified above are intended for use as replacements for existing installations only.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Snubber classification variables (see 1.2).
(c) If union nut, tail piece and O-ring shall be furnished with each snubber that is intended for pressure gauge general applications (see 3.3.2.1.1).
(d) Levels of preservation and packing (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.4	Drawings, engineering and associated list	DI-E-7031	Level 3 Design activity designation - Contractor Drawing numbers - Contractor Delivery of hard copies - Contracting activity
4.1.2	Inspection system program plan	DI-R-4803	----
4.1.2	Calibration system description	DI-R-7064	----
4.3	Report, test/inspection	UDI-T-23473	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., Vol. I, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting activity.)

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6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-2940 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Definitions.

6.5.1 Fluctuation. A nonperiodic, low amplitude (less than 50 percent of pressure sensing instrument span, peak to peak) pressure change.

6.5.2 Pulsation. A periodic, low amplitude (less than 50 percent of pressure sensing instrument span, peak to peak) pressure change.

6.5.3 Spike. A short duration (50 milliseconds or less), high amplitude (minimum 50 percent of pressure sensing instrument span) positive pressure transient.

6.5.4 Snubber. A pressure transmission device which restricts the rate of fluid flow to a pressure sensing instrument and, as a result, the rate of pressure change. The snubber is used in pressure instrument piping systems to suppress pressure spikes and dampen both pressure fluctuations and pulsations before these spikes, fluctuations and pulsations can adversely affect the pressure sensing instrument.

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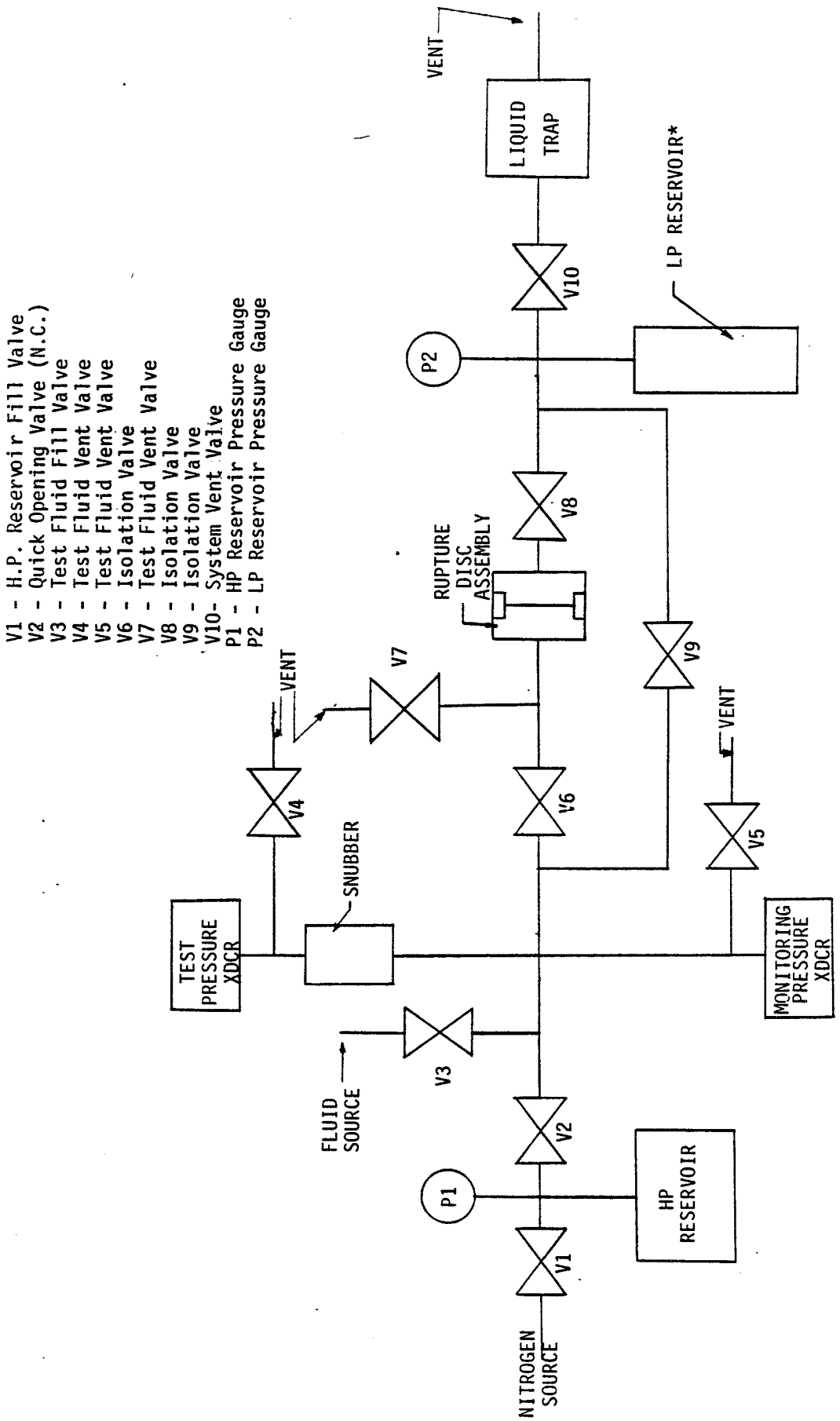
6.6 Subject term (key word) listing.

Fluid pressure
Instrument protection
Pressure gauge
Pressure gauge oxygen
Pressure transducer
Snubbers

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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* Volume of Low Pressure reservoir shall be at least 100 times the volume of the high pressure reservoir.

FIGURE 1. Spike suppression/dampening test fixture.

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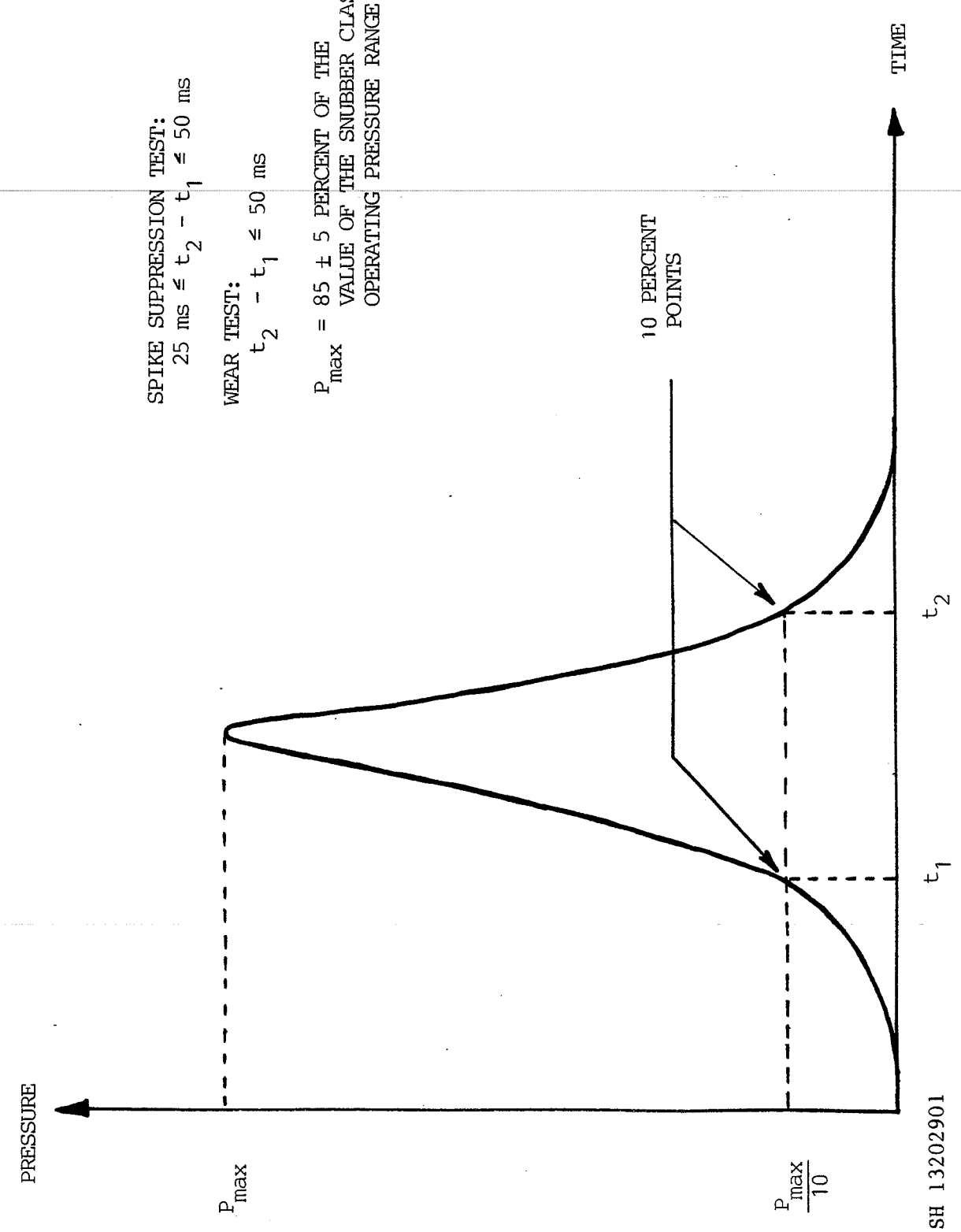


FIGURE 2. Representative pressure spike.

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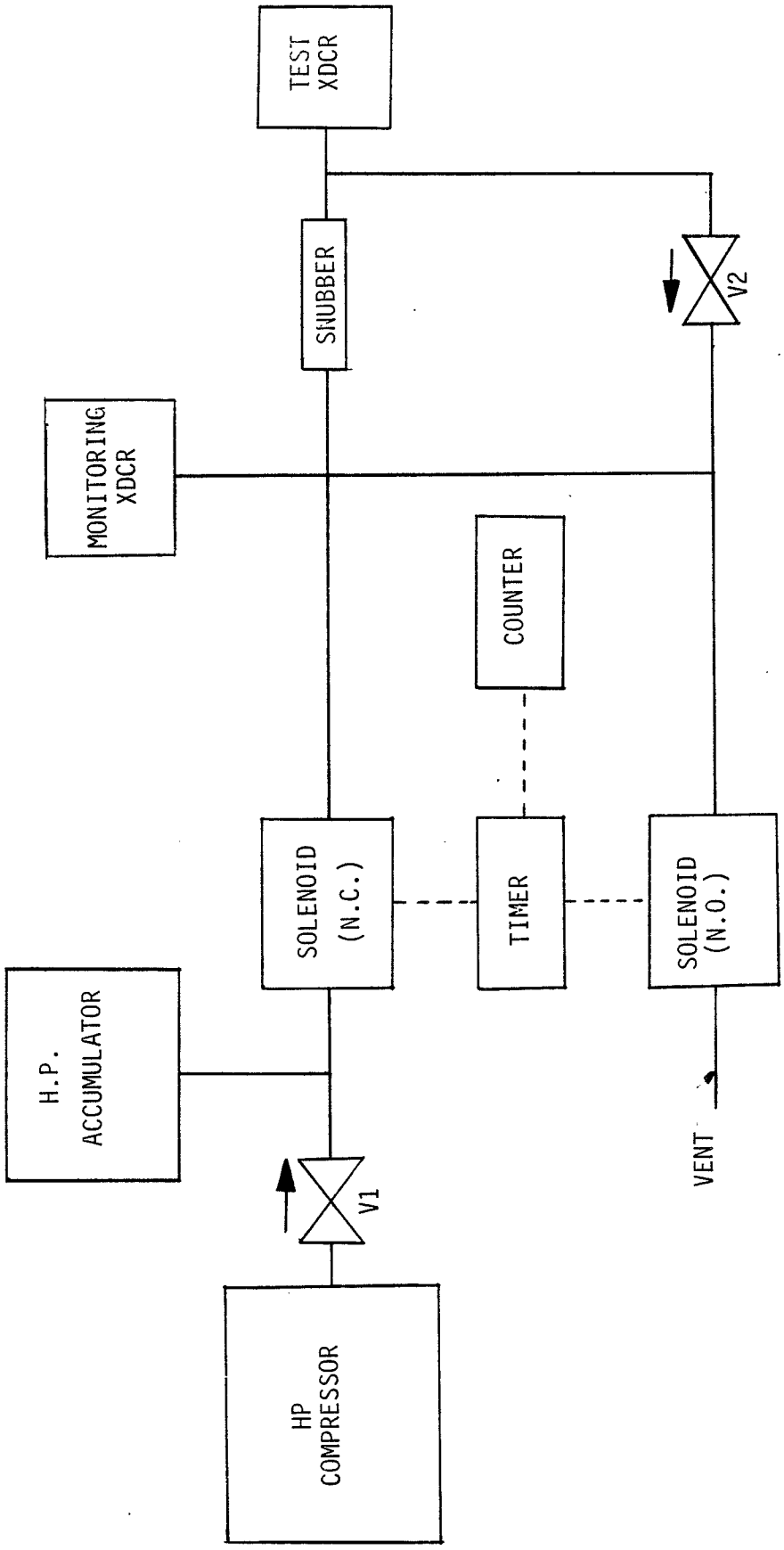


FIGURE 3. Wear test fixture.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-2940B	Snubbers, Fluid Pressure, Instrument Protection

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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